



Health System Management Group Assignment (C2)

Topic: Quantitative and System School of Thought of Management

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Quantitative School of Thought of Management

Historical Background

Along with the ground-breaking shift in information technology application came the quantitative School of thought: a school of ideas based on decision theory, the use of statistics in decision-making and the development of mathematical/econometric models supported by computer technology. This strategy rests on the premise that mathematical methods aid the manager in solutions of issues.

The quantitative management school developed as a methodical means of decision-making. Heavily impacted by World War II military tactics, advancements in the middle of the 20th century mathematics and the development of computer technology, its progress can be followed across three steps essential phases:

1. Early Foundations (pre- 20th Century)

Frederick Taylor, 1911: Though not strictly numerical, Taylor's work introduced time-motion studies and efficiency measures, preparing the groundwork for data-driven management.

Control charts were developed to monitor statistical quality control in the 1920s by Walter Shewhart, later becoming Six Sigma from industrial processes.

2. World War II and Operations Research (1940s–1950s)

The military required best methods for resource distribution, bombing precision, and logistics.

Among mathematicians like John von Neumann (Game Theory) and George Dantzig (Linear Programming created models to address difficult war-related issues.

Businesses started using Operations Research (OR) methods for inventory management after the war.

3. Computer Revolution & Modern Management Science (1960s–present)

Large-scale data processing became possible thanks to computers' emergence, which also enhanced quantitative techniques more.

Emerging Management Information Systems (MIS) incorporated data analysis into decision-making.

Lean manufacturing and total quality management (TQM) integrated statistical techniques for improvement in process.

Big Data, artificial intelligence, and machine learning have today transformed quantitative management into Automation and predictive analysis.

The Quantitative School transformed management into a science-based discipline, reducing reliance on intuition and emphasizing optimization, forecasting, and risk analysis. While

criticized for sometimes neglecting human factors, it remains essential in finance, logistics, healthcare, and technology-driven industries.

Modus Operandi of the Quantitative School of Management Thought

The Quantitative School of Management uses numbers, data, and logical tools to solve real problems. In healthcare, where decisions affect lives and resources are often limited, this approach helps managers make informed, effective choices.

1. Operations Research (OR) and Optimization Models

These techniques use mathematical modeling to solve complex resource allocation problems. It helps hospitals use what they have: staff, time, space, in the most efficient way. For example, using linear programming, a hospital can schedule surgeries to reduce operating room downtime. Queuing theory helps analyze patient flow, so emergency departments can reduce waiting times by adjusting staff levels.

2. Statistical Decision Analysis

This approach incorporates probability theory and risk assessment tools to evaluate uncertain outcomes. With tools like Monte Carlo simulations, hospitals can forecast patient admissions during flu season and plan resources ahead of time. Decision trees help visualize choices like investing in new equipment by comparing risks and benefits.

3. Simulation and Forecasting

Simulation and forecasting techniques, like discrete event modeling and time series analysis, let managers test ideas before applying them. Using models, hospitals can predict future trends like rising admissions or drug shortages. This helps with long-term planning and avoiding crisis decisions.

4. Quality Control and Process Improvement

These methodologies bring statistical rigor to healthcare delivery. Techniques like Six Sigma and Statistical Process Control (SPC) help reduce medical errors, standardize care, and improve patient outcomes.

5. Data driven performance metrics

Performance measurement systems like balanced scorecards and KPIs track financial, clinical, and operational metrics, enabling benchmarking and targeted improvements. These measure things like infection rates, patient satisfaction, and cost per case making sure everyone is aligned with goals.

6. Management Information Systems (MIS)

It integrates quantitative approaches with modern technology. Tech-driven analytics enhance decision-making with real-time predictions and visualized data. They turn raw data into useful dashboards and reports, helping managers respond quickly and accurately.

Strengths of the Quantitative School of Management

1. Scientific and Objective: Decisions are based on data and logic, not guesswork.
2. Precision and Accuracy: Uses mathematical models for clear, exact solutions.
3. Handles Complex Problems: Solves large-scale, multi-variable business issues.
4. Improves Decision-Making: Offers tools like forecasting, simulations, and optimization.
5. Supports Strategic Planning: Helps in long-term planning through models and scenarios.
6. Optimizes Resources: Ensures efficient use of time, money, and manpower.
7. Standardized and Repeatable: Models can be used across industries for consistent results.
8. Enables Performance Measurement: Sets clear benchmarks and tracks progress.
9. Fits with Modern Technology: Supports data systems, AI, and automation tools.
10. Reduces Bias: Promotes rational, evidence-based decisions.
11. Cross-Disciplinary: Combines math, stats, IT, and economics.
12. Widely applicable: Useful in business, healthcare, military and government.

Drawbacks of quantitative school of thought in management

1. Overemphasis on Quantifiable Data: The Quantitative School prioritizes measurable variables (e.g., costs, output, efficiency) while neglecting qualitative factors such as employee morale, leadership style, and corporate culture.

2. Unrealistic Assumptions in Models: Quantitative models often rely on simplistic assumptions (e.g., perfect information, stable demand, linear relationships) that rarely hold in real business environments.

3. High Cost & Complexity of Implementation: Advanced quantitative techniques require: Specialized expertise (statisticians, data scientists), Expensive software (MATLAB, SAS, Python) and Time-consuming model calibration

4. Neglect of Human Judgment & Creativity: The school undervalues managerial intuition, innovation, and strategic vision elements that cannot be quantified but drive long-term success.

5. Ethical & Behavioral Blind Spot: Models optimize for efficiency/profit but often ignore ethical implications, biases, and social responsibility.

6. Inflexibility in Dynamic Environments: Quantitative models rely on historical data, making them ineffective in rapidly changing markets (e.g., tech, pandemics).

7. Short-Term Focus over Long-Term Strategy: Models prioritize immediate KPIs (quarterly profits, cost-cutting) over sustainable growth.

Case Scenarios on Quantitative School of Thought of Management

Case Scenario 1

Adama General Hospital & Medical College is facing a surge in emergency room (ER) visits. Patients report long wait times, and staffs are struggling to efficiently manage resources. The hospital's administration wants to apply quantitative management techniques to make data-driven decisions and streamline operations.

Problem Statement

ER congestion is leading to:

- An average patient waiting time of 2 hours
- Staff overtime costs increasing by 30%
- Patient satisfaction scores dropping below 50%

Quantitative Approach

The hospital administrator partners with the operations research team to address the problem using quantitative tools:

1. Queuing Theory

- Analyze patient arrival patterns and service rates
- Model ER as a multi-server queuing system
- Use formulas to calculate optimal staffing during peak hours

2. Linear Programming

- Allocate resources (doctors, nurses, beds) under constraints (budget, staff availability)
- Maximize efficiency and reduce total operational cost

3. Forecasting Techniques

- Use historical patient arrival data to predict future demand

- Adjust resource allocation based on seasonal trends (e.g. flu season)

Outcome

- After few months of implementing the model:
- Average wait time reduced.
- Overtime costs reduced.
- Patient satisfaction increased.
- Staff reported less burnout and improved workflow.

Summary

By integrating data, statistics, and mathematical models, the quantitative approach fundamentally transforms organizational management. It empowers managers to move beyond intuition, gain deeper insights, and make informed decisions across all functions which in turn results in; optimized resource allocation, enhanced strategic planning, a culture of continuous improvement, and a competitive advantage through superior performance and innovation.

The System School of thought of management

Introduction

This theory is based on the assumption that an organization is a system and its components are inter-dependent and inter-related. The effectiveness of system is largely dependent upon the inter-relations and interdependency of the different sub-systems. The organization is responsive to environmental influence.

Historical Background

Ludwig Von Bertalanffy systematized General Systems Theory (GST) in the 1930s, proposing that general principles can be applied to all systems biological, mechanical, social. Cybernetics by Norbert Wiener (1948) included feedback loops, crucial to control systems and early computing. These contributed to the view of systems as interconnected, adaptable, and self-regulating.

In Mid-20th Century Formalization: GST influenced interdisciplinary research, and operations research applied systemic methods to business and the military. The Tavistock Institute integrated social and technical system analysis in industry. Jay Forrester established system dynamics, feedback and causal loop modeling (e.g., city planning). And Stafford Beer established management cybernetics and the Viable System Model (VSM) for organizational health. It was superseded in the 1970s– 1980s by Peter Checkland's Soft Systems Methodology (SSM) to handle ill-defined social problems through participatory stakeholder engagement. And Donella Meadows' Limits to Growth (1972) warned against unbridled growth with systems models that are still widely referenced in environmental and sustainability projects. Then from the late 20th Century to the Present, Complexity science met self-organizing systems (e.g., Financial markets) with institutions Critical systems thinking, led by Michael Jackson and

Robert Flood, emphasized ethics, diversity, and power relationships in system interventions (e.g., inclusion of Indigenous knowledge in governance).

System means a set of elements standing in interrelation among them and with the environment. The most important ones are the joint action of the components to generate a whole and dynamic system. An open system, 'interacts with its environment'. If the system is decomposed, it can be found that it is composed of various sub-systems. Similarly, an organization is also one sub-system of a larger environment. It acknowledges the impact of the external world on all that happens within an organization. It provides a rigorous conceptual model for good analysis and understanding of organizations. It is normally closer to reality

Key Characteristics of Systems School of Management Thought

Subsystem integration: An organization is a system consisting of several sub-systems which are mutually related to each other. All sub-systems should be studied in relation to each other rather than separately. Effective management ensures coordination among all parts.

Holistic view: A system consists of interacting elements. It is a set of inter-related and inter-dependent parts arranged in a manner that produces a unified whole. Decisions made in any subsystem affect the entire system.

Open Vs. closed system: A system can be open or closed. An open system interacts with its environment. Traditional organization theorists viewed organizations as closed systems, while modern perspectives see them as open systems that continually interact with their environment, including external forces like government agencies, customers, and suppliers.

Input-process-output: Every system has flows of information, material and energy and these inputs get converted into outputs of goods, services and satisfactions in the organization.

Synergy: The collective output of the whole system is greater than the sum of the outputs of its sub-systems. Collaboration among units leads to better results.

Goal orientation: All sub-systems must focus their efforts on achieving the organization's goals.

System boundary: An organizational system has a boundary that defines what parts are internal and what parts are external, separating the system from its environment. This boundary can be physical, operational, or conceptual.

Feedback mechanism: The organization responds to environmental effects. A system adjusts and reacts to changing conditions in its surroundings, maintaining control over its operations through feedback. Continuous feedback helps the organization adjust and improve.

Advantages of Systems School of Management Thought

Holistic Perspective: Instead of focusing on isolated parts, the systems approach encourages managers to see the organization as a whole.

Better Understanding of Complex Problems: provides a framework for analyzing complex problems by considering the various factors and relationships involved.

Promotes Efficiency and Effectiveness: By optimizing all parts of the system to work together, organizations can reduce waste and improve output quality which helps in enhancing of resource utilization.

Improved Coordination and Communication: By emphasizing the interconnectedness of different departments and functions it improves coordination and communication.

Focus on Interdependence: Emphasizes that all departments (subsystems) are connected. Also encourages collaboration between units.

Enhanced Adaptability and Flexibility: recognizes that organizations operate in a dynamic environment and must be able to adapt to change.

Supports Continuous Improvement: Feedback loops encourage learning and correction also helps in mistakes or inefficiencies identification and correction systematically.

Improved Long Term Planning: Because the systems approach considers the long term effects of decisions, it can help managers make more sustainable and strategic plans.

Applicable across Industries: The systems approach is flexible and can be applied to: Business, Healthcare, Education, Government and Non profits

Limitations of systems school of management thoughts

1. Complexity and Abstraction

Complexity: Systems thinking can lead to the formation of complex plans that are difficult to understand and change them into actions. The managers can sometimes be overwhelmed with too much information because of the focus on interrelationships and feedback.

Abstraction: when trying to understand the whole system, the focus can shift from the practical, day to day actions to abstract concepts, which makes it hard changing it into actions.

2. Lack of Specificity

It often provides general principles but it lacks specific, step by step instructions for answering problems in the management.

3. Implementation Challenges

It requires managers to have a broad understanding of all parts of the organization with their interconnections, which is difficult in large or rapidly changing environments.

It requires significant investment in training, technology, and large data analysis, which is difficult for small or resource limited organizations. This makes it expensive and time consuming.

4. Too much focus on Stability

Focusing on maintaining stability can sometimes prevent adaptation to changing environments.

5. Too much focus on Interdependence

Since the system approach may place too much focus on relationships between subsystems, it can neglect the importance of individual performance. This may ignore the role of individual motivation, emotions, values and informal relationships, which are important to the work place.

Systems thinking can lead to analysis paralysis when managers become more concerned with every potential interaction instead of taking action.

Case Scenario on Systems School of Thought of Management

One of the hospitals in Addis Ababa is facing several issues like:

Delayed patient service

Frequent deficiency and stock out in essential medication and Frequent interruption in power supply which even led to dysfunction in many essential machines.

Frequent Interruption in water supply

Despite, from outside, the hospital having good infrastructure and relatively reasonable amount of skilled health workers, it isn't performing optimally. Because of this dire condition, Dr. Selam, an excellent medical director, is assigned as a new head of hospital to handle the situation.

If Dr. Selam decided to use Modus operandi of systems school of thought, how would her framework would look like?

Step 1: Understanding “ What type of system is the hospital? “

She acknowledged the hospital as an ***open system*** which has many interrelated sub system (Medical departments, Nursing, Pharmacy, Radiology, HR and Finance etc.) in it, and also constantly interacts with external environment (Patients, Government, Non health sector infrastructures (ELPA etc...))

Hence, foresaw possible she might come in contact with.

Step 2: Identifying Inputs, Processes, and Outputs

- Inputs: Patients, Medical Supplies, Staff, Water, Electricity, Funding
- Processes: Triage, Diagnosis, Treatment, Reporting, Follow up
- Output: Cured patients, Dead patients, Referred patients, Administered vaccines

After recognition of the hospital as an open system which has many ***interrelated/ interdependent*** sub- system in it and which constantly interacts with the outside Dr. Selam, as per stated in

Systems SOT , knew that the only way to change the current condition is to integrate all subsystem together and amass the “ **advantage of synergy** “ by solving the problem of each sub-system and **establishing a constant feedback track** within the hospital and also with the outside to sustain the fore brought changes and tackle the future problem before they get big.

Step 3: Establishing and Enhancing Feedback Loops

She initiated regular **feedback from patients and staff**

- Putting the manual feedback book in different sections of the hospital for the patients and staff
- Establishing a new room with a digital recording / one personnel who takes a complaint from patients and attendants who can't write
- Adding a feedback section on the hospital's Website for those who want to file their complaint online.
- Monthly meetings to evaluate performance.

Step 4: Subsystem & Inter - Sectorial Integration

The major problem for the patient delay in the hospital was the reason that every clinical (**Lab** referrals, Imaging referrals, prescriptions etc.) activities are done manually (via paper) and there is no system to integrate different departments of hospital in real time.

So Dr. Selam ordered for **integrated inventory software called “ iCARE”** to be installed in the hospital and training for all staffs to be given.

She also held a meeting with ELPA and discussed any specific measures that could be done to alleviate the power interruption. And also start to search for donors / governmental subsidies that would buy “ Standby Generator “ for the hospital. She held a meeting with the hospital pharmacy staff and different pharmaceutical P.L.C.s and put out a bidding for them to participate on.

The water problem was all across the city and can't be solved soon. Hence, Dr. Selam ordered four water storage aquariums with the sanitizing system to be bought and installed in the hospital.

Outcome:

- Patient wait times decrease.
- Medicine availability improves.
- The hospital got a donor; a standby generator is bought and power problem is solved. So is the water problem.
- Staff morale rises.

Key Systems Concepts in this Case:

Concept
Subsystems
Interdependence
Open System
Feedback
Synergy

Summary

The systems school of management emphasizes achieving organizational goals through its components and their dynamic interaction, viewing the organization as an integrated whole.

It explains how organizational problems can be better understood by analyzing the relationships and patterns among different components rather than viewing them in isolation.

It is beneficial for effective decision making and increasing operational efficiency since all its subsystems work in harmony.

It comprises of inputs, transformation processes, outputs and feedbacks to better tackle organizational concerns.

- ✓ Overall it continues to influence modern management practices by providing a comprehensive framework in which an organization can function effectively.

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